AthDGC, Athens Digital Glossa Chronos (Athens-PROIEL)

An open PROIEL-style treebank with Indo-European parallels

Nikolaos Lavidas

Background

Greek is the longest continuously attested member of the Indo-European family. The PROIEL (the dependency-treebank (a collection of sentences whose grammatical structure has been analysed and stored) standard for early Indo-European languages, developed at Oslo) Treebank (Haug & Jøhndal 2008; Eckhoff *et al.* 2018) established the gold standard for syntactically annotated parallel IE Bible corpora, but its scope is restricted to the New Testament. No openly licensed end-to-end workflow currently covers the full diachronic record of Greek with explicit argument-structure tagging and reproducible cross-lingual alignment (matching corresponding words or sentences between texts in different languages).

Approach

AthDGC ("Athens-PROIEL") is an open end-to-end workflow that:

1. **Discovers** new Greek + parallel-language sources daily (archive.org, Perseus, First1K, Wikisource, Diorisis)
2. **Filters** via Greek-script ratio, apparatus-criticus rejection, content-hash dedup
3. **Converts** to PROIEL XML 2.0 (the file format that stores each sentence as a tree of word-by-word grammatical relations, developed at Oslo)
4. **Annotates** with Stanza (Stanford's open-source Python workflow that automatically tags, lemmatises, and parses sentences) `grc_proiel` (+ `la_proiel`, `cu_proiel`, `got_proiel` for parallels)
5. **Extracts** per-verb argument structure (subj, obj, oblique, voice, aspect)
6. **Aligns** cross-lingually via LaBSE (a multilingual sentence-embedding model that maps sentences from many languages into a common vector space) (sentence) + AwesomeAlign (a word-alignment procedure that uses multilingual-BERT attention to match words across translations) (word, mBERT)
7. **Stores** as PROIEL XML + JSONL (a plain-text data format that stores one record per line) partitions + Qdrant vectors + Neo4j (a graph database that stores data as nodes and connections rather than as tables) alignment graph (a database that records which token in language A corresponds to which token in language B)

Coverage (v0.4)

- 8 diachronic Greek periods (Archaic, Modern)
- 4 cross-aligned IE witnesses (Latin, Gothic, OCS, Classical Armenian in ingestion)

- TLG canonical-author metadata curated
- Verified scale ships with v0.5 release notes (post-ARIS audit)

Argument-structure capture

Per VERB / AUX token AthDGC stores:

- subject, direct object, indirect / oblique args
- voice (active / middle / passive)
- aspect (perfective / imperfective)
- tense, mood, verb-form, person, number

stored in a column queryable across the cross-lingual alignment edges.

Example query

> "Show every Greek aorist transitive verb with an accusative object whose Latin Vulgate counterpart is a passive periphrastic"

Neo4j Cypher over `(:Token)-[:TRANSLATED_AS]->(:Toke` edges, filtering source by `(voice=Act,aspect=Perf,obj=Acc)` and target by `(voice=Pass,verb_form=Periphrasti`

Open access

- Source code (Apache-2.0 (a permissive open-source licence widely used for software)): github.com/AthDGC/Diachronic-Linguistics-Platform
- Dataset (CC-BY-4.0 (the Creative Commons Attribution 4.0 licence, which permits reuse with credit)): 10.5281/zenodo (an open research-data repository hosted at CERN that mints permanent DOIs).20439182
- Showcase: athdgc.github.io
- HuggingFace: athdgc/diachronic-greek-corpus (forthcoming)

Funding

Funded by the **Hellenic Foundation for Research and Innovation (HFRI)**, 3rd Call for HFRI Research Projects to support Post-Doctoral Researchers, **Project No. 20577**; with support from **Greece 2.0 NRRP**. Project: **CVL-CDSAML** at NKUA. Compute: GRNET ARIS (the Greek national high-performance computing cluster, run by GRNET) (`pa260305`).